

ASSIGNMENT - 1

1) CREATING WAREHOUSES FOR Marketing Analysts, Finance Team, ETL Developers

I) CREATING WAREHOUSE FOR MARKETING ANALYSTS

QUERY:

```
CREATE OR REPLACE WAREHOUSE MARKETING_WH
```

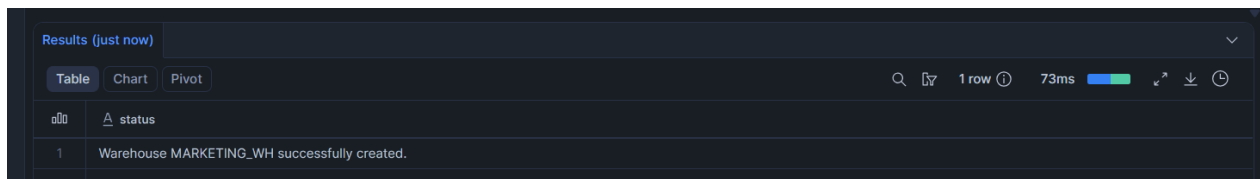
```
WITH
```

```
WAREHOUSE_SIZE = 'SMALL'
```

```
WAREHOUSE_TYPE = 'STANDARD'
```

```
AUTO_SUSPEND = 300
```

```
AUTO_RESUME = TRUE
```



The screenshot shows a SQL query results window with a dark theme. At the top, it says "Results (just now)". Below this are tabs for "Table", "Chart", and "Pivot". To the right of the tabs, there is a search icon, a refresh icon, and text indicating "1 row" and "73ms". Below the tabs, there is a table with one row. The first column is labeled "id" and contains the value "1". The second column is labeled "status" and contains the text "Warehouse MARKETING_WH successfully created.".

id	status
1	Warehouse MARKETING_WH successfully created.

ii) CREATING WAREHOUSE FOR FINANCE TEAM

QUERY:

```
CREATE OR REPLACE WAREHOUSE FINANCE_WH
```

```
WITH
```

```
WAREHOUSE_SIZE = 'XSMALL'
```

```
WAREHOUSE_TYPE = 'STANDARD'
```

```
AUTO_SUSPEND = 300
```

```
AUTO_RESUME = TRUE
```

28 WAREHOUSE_TYPE = 'STANDARD'

Results (just now)

Table Chart Pivot

1 row 76ms

status
Warehouse FINANCE_WH successfully created.

iii) CREATING WAREHOUSE FOR ETL DEVELOPERS

QUERY :

CREATE OR REPLACE WAREHOUSE ETL_WH

WITH

WAREHOUSE_SIZE = 'MEDIUM'

WAREHOUSE_TYPE = 'STANDARD'

AUTO_SUSPEND = 300

AUTO_RESUME = TRUE

Results (just now)

Table Chart Pivot

1 row 78ms

status
Warehouse ETL_WH successfully created.

IV) DISPLAYING ALL WAREHOUSES

SHOW WAREHOUSES;

Results (just now)

Table Chart Pivot

5 rows 65ms

	name	state	type	size	# running	# queued	is_default	is_current	auto_suspend	auto_resume
1	COMPUTE_WH	SUSPENDED	STANDARD	X-Small	0	0	Y	N	600	true
2	ETL_WH	STARTED	STANDARD	Medium	0	0	N	Y	300	true
3	FINANCE_WH	STARTED	STANDARD	X-Small	0	0	N	N	300	true
4	MARKETING_WH	STARTED	STANDARD	Small	0	0	N	N	300	true
5	SYSTEM\$STREAMLIT	SUSPENDED	STANDARD	X-Small	0	0	N	N	60	true

2) CREATING ROLE FOR MARKETING ANALYST, FINANCE TEAM, ETL DEVELOPERS

CREATE OR REPLACE ROLE MARKETING_ANALYST;

CREATE OR REPLACE ROLE FINANCE_USER;

CREATE OR REPLACE ROLE ETL_DEVELOPER;

DISPLAYING ALL THE ROLES



The screenshot shows a database query results window with a table of roles. The table has columns for row number, created_on, name, is_default, is_current, is_inherited, assigned_to_users, granted_to_roles, and granted_roles. The data is as follows:

	created_on	name	is_default	is_current	is_inherited	assigned_to_users	granted_to_roles	granted_roles
1	2026-05-12 21:59:46.041 -0700	ACCOUNTADMIN	Y	Y	N	1	0	
2	2026-05-15 05:10:42.767 -0700	ETL_DEVELOPER	N	N	N	0	0	
3	2026-05-15 05:10:35.500 -0700	FINANCE_USER	N	N	N	0	0	
4	2026-05-15 05:10:28.477 -0700	MARKETING_ANALYST	N	N	N	0	0	
5	2026-05-12 21:59:46.368 -0700	ORGADMIN	N	N	N	1	0	

i) GRANT ACCESS TO THE ROLES

GRANT USAGE ON WAREHOUSE MARKETING_WH TO ROLE MARKETING_ANALYST;

GRANT USAGE ON WAREHOUSE FINANCE_WH TO ROLE FINANCE_USER;

GRANT USAGE ON WAREHOUSE ETL_WH TO ROLE ETL_DEVELOPER;

3) SETTING UP USERS WITH ROLES

i) CREATING MARKETING ANALYST

USER CREATE OR REPLACE USER

MKT_USER1 PASSWORD =

'MKTUSER1'

LOGIN_NAME = 'MKT_USER1'

DEFAULT_ROLE = 'MARKETING_ANALYST'

DEFAULT_WAREHOUSE = 'MARKETING_WH'

MUST_CHANGE_PASSWORD = FALSE;

II) CREATING FINANCE TEAM USER

```
CREATE OR REPLACE USER FIN_USER1  
  
PASSWORD = 'FINUSER1'  
  
LOGIN_NAME = 'FIN_USER1'  
  
DEFAULT_ROLE = 'FINANCE_USER'  
  
DEFAULT_WAREHOUSE = 'FINANCE_WH'  
  
MUST_CHANGE_PASSWORD = FALSE;
```

III) CREATING ETL DEVELOPER USER

```
CREATE OR REPLACE USER ETL_USER1  
  
PASSWORD = 'ETLUSER1'  
  
LOGIN_NAME = 'ETL_USER1'  
  
DEFAULT_ROLE = 'ETL_DEVELOPER'  
  
DEFAULT_WAREHOUSE = 'ETL_WH'  
  
MUST_CHANGE_PASSWORD = FALSE;
```

IV) GRANT THE ROLES TO USERS

GRANT ROLE MARKETING_ANALYST TO USER MKT_USER1;

GRANT ROLE FINANCE_USER TO USER FIN_USER1;

GRANT ROLE ETL_DEVELOPER TO USER ETL_USER1;

DISPLAYING ALL THE USERS WITH THEIR ROLES THAT ASSIGNED

Results (3 minutes ago)

	name	created_on	login_name	display_name	first_name	last_name	email	mins.
1	ETL_USER1	2026-05-15 05:19:15.284 -0700	ETL_USER1	ETL_USER1	null	null	null	null
2	FIN_USER1	2026-05-15 05:19:08.002 -0700	FIN_USER1	FIN_USER1	null	null	null	null
3	MKT_USER1	2026-05-15 05:19:01.890 -0700	MKT_USER1	MKT_USER1	null	null	null	null
4	RAMJAN	2026-05-12 21:59:45.974 -0700	RAMJAN	RAMJAN	Kasi Ramjan Vali	Dudekula	dudekulakasiramjanvali	null

4) Create Performance Demo Database

i) CREATING DATABASE

CREATE OR REPLACE DATABASE

RETAIL_MARKETING_DB;

USE DATABASE RETAIL_MARKETING_DB;

USE SCHEMA PUBLIC;

ii) CREATE CAMPAIGN_SALES TABLE

CREATE OR REPLACE TABLE CAMPAIGN_SALES AS

SELECT

SS.SS_SOLD_DATE_SK,

DD.D_DATE,

SS.SS_CUSTOMER_SK,

C.C_FIRST_NAME,

C.C_LAST_NAME,

SS.SS_ITEM_SK,

I.I_ITEM_DESC,

SS.SS_QUANTITY,

SS.SS_SALES_PRICE,

SS.SS_NET_PROFIT

FROM SNOWFLAKE_SAMPLE_DATA.TPCDS_SF10TCL.STORE_SALES

SS JOIN SNOWFLAKE_SAMPLE_DATA.TPCDS_SF10TCL.CUSTOMER

C

ON SS.SS_CUSTOMER_SK = C.C_CUSTOMER_SK

JOIN

SNOWFLAKE_SAMPLE_DATA.TPCDS_SF10TCL.DATE_DIM DD

ON SS.SS_SOLD_DATE_SK = DD.D_DATE_SK

JOIN

SNOWFLAKE_SAMPLE_DATA.TPCDS_SF10TCL.ITEM I

ON SS.SS_ITEM_SK = I.I_ITEM_SK

LIMIT 10000

5)Grant Permissions

i) GRANT DATABASE ACCESS

```
GRANT USAGE ON DATABASE RETAIL_MARKETING_DB  
TO ROLE MARKETING_ANALYST;  
  
GRANT USAGE ON DATABASE RETAIL_MARKETING_DB  
TO ROLE FINANCE_USER;  
  
GRANT USAGE ON DATABASE RETAIL_MARKETING_DB  
TO ROLE ETL_DEVELOPER;
```

II) GRANT SCHEMA ACCESS

```
GRANT USAGE ON SCHEMA RETAIL_MARKETING_DB.PUBLIC  
TO ROLE MARKETING_ANALYST;  
  
GRANT USAGE ON SCHEMA RETAIL_MARKETING_DB.PUBLIC  
TO ROLE FINANCE_USER;  
  
GRANT USAGE ON SCHEMA RETAIL_MARKETING_DB.PUBLIC  
TO ROLE ETL_DEVELOPER;
```

III) Grant Read Access to Marketing & Finance

```
GRANT SELECT ON TABLE CAMPAIGN_SALES  
TO ROLE MARKETING_ANALYST;  
  
GRANT SELECT ON TABLE CAMPAIGN_SALES  
TO ROLE FINANCE_USER;
```

IV) Grant Full Access to ETL Developer

GRANT ALL PRIVILEGES ON TABLE

CAMPAIGN_SALES TO ROLE ETL_DEVELOPER;

v) DISPLAYING THE GRANTS ON THE TABLE

SHOW GRANTS ON TABLE CAMPAIGN_SALES;

	created_on	privilege	granted_on	name	granted_to	grantee_name
	5/15/2026	SELECT 23.1% APPLYB... 7.7% +9 more	TABLE 100.0%	RETAIL_MARKETING_DB.PUBLIC.CAMPAIGN_S...	ROLE 100.0%	ETL_DEVEL... 76.9% ACCOUNTA... 7.7% +2 more
1	2026-05-15 05:58:16.021 -0700	OWNERSHIP	TABLE	RETAIL_MARKETING_DB.PUBLIC.CAMPAIGN_SALES	ROLE	ACCOUNTADMIN
2	2026-05-15 08:41:39.419 -0700	APPLYBUDGET	TABLE	RETAIL_MARKETING_DB.PUBLIC.CAMPAIGN_SALES	ROLE	ETL_DEVELOPER
3	2026-05-15 08:41:39.391 -0700	DELETE	TABLE	RETAIL_MARKETING_DB.PUBLIC.CAMPAIGN_SALES	ROLE	ETL_DEVELOPER
4	2026-05-15 08:41:39.368 -0700	EVOLVE SCHEMA	TABLE	RETAIL_MARKETING_DB.PUBLIC.CAMPAIGN_SALES	ROLE	ETL_DEVELOPER
5	2026-05-15 08:41:39.412 -0700	INSERT	TABLE	RETAIL_MARKETING_DB.PUBLIC.CAMPAIGN_SALES	ROLE	ETL_DEVELOPER
6	2026-05-15 08:41:39.426 -0700	REBUILD	TABLE	RETAIL_MARKETING_DB.PUBLIC.CAMPAIGN_SALES	ROLE	ETL_DEVELOPER
7	2026-05-15 08:41:39.384 -0700	REFERENCES	TABLE	RETAIL_MARKETING_DB.PUBLIC.CAMPAIGN_SALES	ROLE	ETL_DEVELOPER
8	2026-05-15 08:41:39.398 -0700	SELECT	TABLE	RETAIL_MARKETING_DB.PUBLIC.CAMPAIGN_SALES	ROLE	ETL_DEVELOPER
9	2026-05-15 08:41:39.360 -0700	SELECT ERROR TAB	TABLE	RETAIL_MARKETING_DB.PUBLIC.CAMPAIGN_SALES	ROLE	ETL_DEVELOPER
10	2026-05-15 08:41:39.377 -0700	TRUNCATE	TABLE	RETAIL_MARKETING_DB.PUBLIC.CAMPAIGN_SALES	ROLE	ETL_DEVELOPER
11	2026-05-15 08:41:39.405 -0700	UPDATE	TABLE	RETAIL_MARKETING_DB.PUBLIC.CAMPAIGN_SALES	ROLE	ETL_DEVELOPER
12	2026-05-15 08:40:41.334 -0700	SELECT	TABLE	RETAIL_MARKETING_DB.PUBLIC.CAMPAIGN_SALES	ROLE	FINANCE_USER
13	2026-05-15 08:40:36.287 -0700	SELECT	TABLE	RETAIL_MARKETING_DB.PUBLIC.CAMPAIGN_SALES	ROLE	MARKETING_ANALYST

6) PERFORMANCE TESTING

Performance Testing Using XSMALL Warehouse

i) USE WAREHOUSE MARKETING_WH;

ii) ALTER WAREHOUSE MARKETING_WH

iii) SET WAREHOUSE_SIZE = 'XSMALL';

SELECT D_DATE, SUM(SS_SALES_PRICE) AS TOTAL_SALES,

SUM(SS_NET_PROFIT) AS TOTAL_PROFIT

FROM CAMPAIGN_SALES

GROUP BY D_DATE

ORDER BY TOTAL_SALES DESC;

MARKETING_WH

RAMJAN

Analyze with Cortex Code

Query Details

Query Profile

Query Telemetry

Details

Status

Success

Duration

2.3s

Driver Status

N/A

Start Time

5/15/2026, 9:18:33 PM

Query ID

01c46334-0001-baae-000e-bf420002240a

Client Driver

Snowsight

End Time

5/15/2026, 9:18:35 PM

Query Tag

—

Session ID

63338377245

Warehouse Size

X-Small

Attributed Compute Credits

—

Parameterized query hash

3d903520beedb6d04e781f20ab1455d5

Query Details

Query Profile

Query Telemetry

+

—

Result [0]

0%

CAMPAIGN_SALES.D_DATE,SUM(CAMP...

17

Sort [1]

0%

SUM(CAMPAIGN_SALES.SS_SALES_PRI...

17

Aggregate [2]

0.6%

SUM(CAMPAIGN_SALES.SS_SALES_PRI...

100k

TableScan [3]

97.7%

RETAIL_MARKETING_DB.PUBLIC.CAMP...

Query insights

None detected. [Learn more](#)

Most expensive nodes(1 of 4)

TableScan [3] 97.7%

Profile overview(Finished)

Total execution time (2.1s) 100.0%

Processing 0.6%

Remote disk I/O 24.4%

Synchronization 73.3%

Initialization 1.7%

Statistics

Scan progress 100.00%

Bytes scanned 1.57MB

Percentage scanned from cache 0.00%

Partitions scanned 9

Performance Testing Using SMALL Warehouse

i) ALTER WAREHOUSE MARKETING_WH

SET WAREHOUSE_SIZE = 'SMALL';

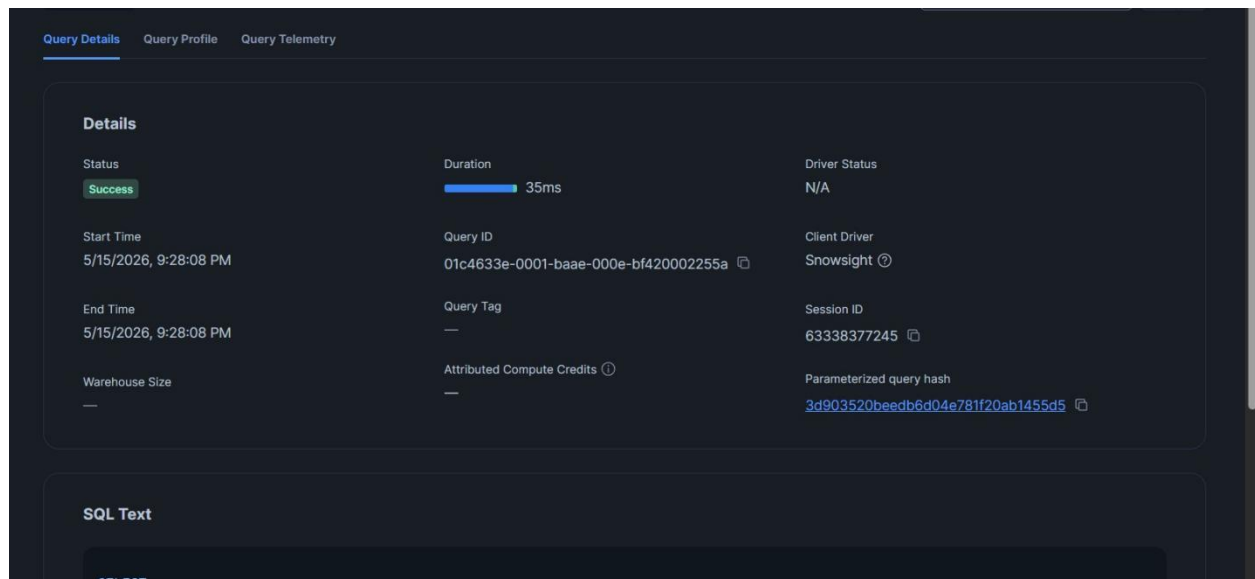
ii) SELECT D_DATE, SUM(SS_SALES_PRICE) AS TOTAL_SALES,

SUM(SS_NET_PROFIT) AS TOTAL_PROFIT

FROM CAMPAIGN_SALES

GROUP BY D_DATE

ORDER BY TOTAL_SALES DESC;

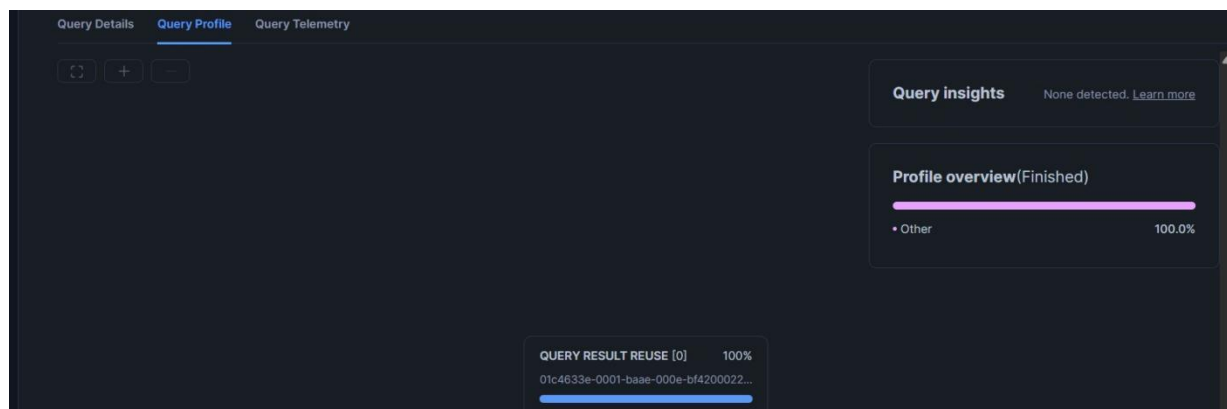


This screenshot shows the 'Query Details' tab in a dark-themed interface. It displays various metadata for a successful query. The 'Details' section is organized into three columns. The first column shows the status as 'Success' with a green indicator, the start time as '5/15/2026, 9:28:08 PM', the end time as '5/15/2026, 9:28:08 PM', and the warehouse size as '—'. The second column shows the duration as '35ms' with a blue progress bar, the query ID as '01c4633e-0001-baae-000e-bf420002255a', the query tag as '—', and the attributed compute credits as '—'. The third column shows the driver status as 'N/A', the client driver as 'Snowsight', the session ID as '63338377245', and the parameterized query hash as '3d903520beedb6d04e781f20ab1455d5'. Below this, the 'SQL Text' section is partially visible, showing the start of the query: 'SELECT'.

Details		
Status	Duration	Driver Status
Success	35ms	N/A
Start Time	Query ID	Client Driver
5/15/2026, 9:28:08 PM	01c4633e-0001-baae-000e-bf420002255a	Snowsight
End Time	Query Tag	Session ID
5/15/2026, 9:28:08 PM	—	63338377245
Warehouse Size	Attributed Compute Credits	Parameterized query hash
—	—	3d903520beedb6d04e781f20ab1455d5

SQL Text

SELECT



This screenshot shows the 'Query Profile' tab in a dark-themed interface. It displays query insights and a profile overview. The 'Query insights' section shows 'None detected' with a link to 'Learn more'. The 'Profile overview' section shows a 'Finished' status with a green progress bar and a single category 'Other' at 100.0%. At the bottom, a 'QUERY RESULT REUSE' section shows '[0]' reuses at 100% for the query ID '01c4633e-0001-baae-000e-bf420002255a'.

Query insights	
None detected.	Learn more

Profile overview(Finished)	
Other	100.0%

QUERY RESULT REUSE [0] 100%

01c4633e-0001-baae-000e-bf420002255a

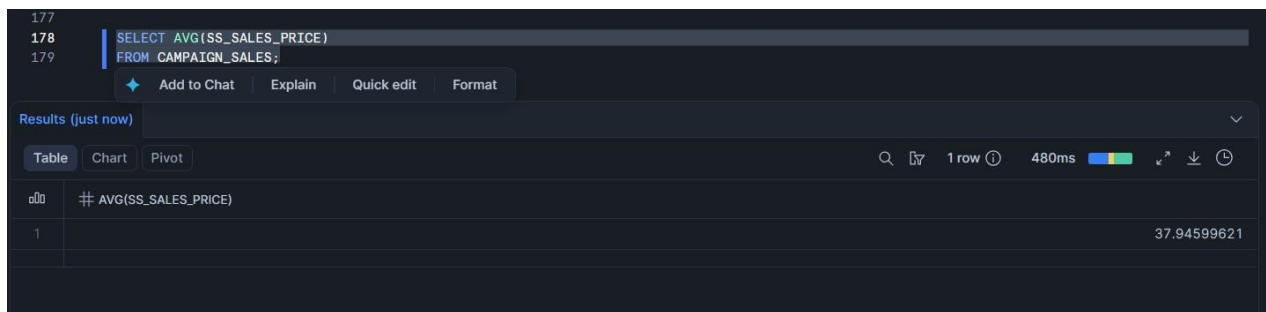
iii) SCALE DOWN AGAIN

```
ALTER WAREHOUSE MARKETING_WH
```

```
SET WAREHOUSE_SIZE = 'XSMALL';
```

7) CACHE TESTING

V) First Query Execution Without Cache



The screenshot shows a SQL query editor with the following text:

```
177  
178 SELECT AVG(SS_SALES_PRICE)  
179 FROM CAMPAIGN_SALES;
```

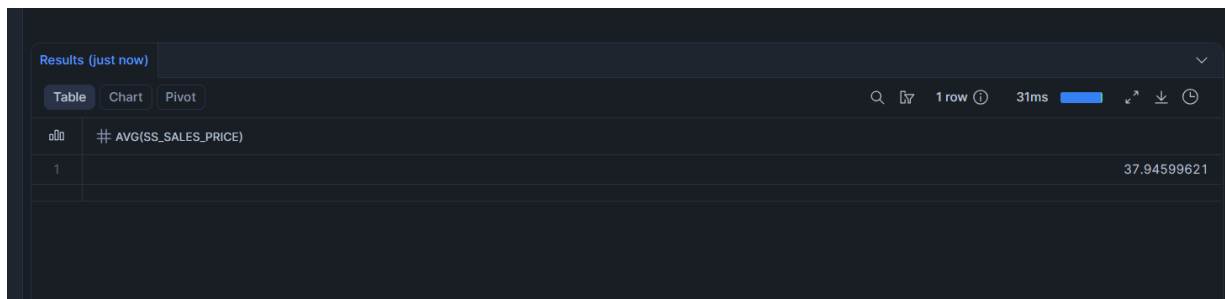
Below the editor, there are buttons: "Add to Chat", "Explain", "Quick edit", and "Format".

The "Results (just now)" section shows a table with the following data:

	# AVG(SS_SALES_PRICE)
1	37.94599621

Metadata: 1 row, 480ms

ii) Second Query Execution Using Query Result Cache



The screenshot shows the same SQL query execution interface as above, but with the following metadata:

Metadata: 1 row, 31ms

	# AVG(SS_SALES_PRICE)
1	37.94599621

8) Cluster Key Testing

i) Query Performance Before Clustering

```
181 //CLUSTER KEY TESTING
182 SELECT
183     D_DATE,
184     SS_SALES_PRICE,
185     SS_NET_PROFIT
186 FROM CAMPAIGN_SALES
187 WHERE D_DATE BETWEEN '1998-12-12' AND '2002-09-29';
188
```

results (just now)

Table Chart Pivot

100,000 rows 636ms

	D_DATE	SS_SALES_PRICE	SS_NET_PROFIT
1	2000-10-09	40.48	37.20
2	2000-10-09	5.00	-1200.54
3	2000-10-09	98.15	258.00
4	2000-10-09	16.27	-674.40
5	2000-10-09	2.86	-35.51
6	2000-10-09	29.60	-1398.96
7	2000-10-09	14.07	-77.00
8	2000-10-09	100.90	2814.45
9	2000-10-09	51.24	-1911.10

Query Performance After Clustering

ADDING CLUSTER KEY

```
180
181 //CLUSTER KEY TESTING
182 SELECT
183     D_DATE,
184     SS_SALES_PRICE,
185     SS_NET_PROFIT
186 FROM CAMPAIGN_SALES
187 WHERE D_DATE BETWEEN '1998-12-12' AND '2002-09-29';
188
189
190 ALTER TABLE CAMPAIGN_SALES
191 CLUSTER BY (D_DATE);
```

Results (just now)

Table Chart Pivot

100,000 rows 322ms

	D_DATE	SS_SALES_PRICE	SS_NET_PROFIT
1	1998-12-12	25.49	-21.96
2	1998-12-12	23.79	-3950.24
3	1998-12-12	10.06	-2029.26
4	1998-12-12	54.82	475.44
5	1998-12-12	6.21	-798.56
6	1998-12-12	20.89	-1069.36
7	1998-12-12	85.30	-35.04
8	1998-12-12	10.10	-3699.31
9	1998-12-12	7.14	-1458.24

